LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 3 trial 1 seed 0

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

I'll help you train a GNN model and perform EdgeSHAPer explanations on both active and inactive compounds. Let me start by finding an available dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some ChemBL activity datasets. Let me load one and start the GNN workflow. 📊

(Calling tool `load_dataset`)

Assistant

Excellent! I have a dataset with 8,348 compounds. Now let me split it for GNN training. 📂

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! 🎯 Now let me prepare the GNN dataset and train the model.

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared! 📊 Now let me train the GNN model.

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN model training submitted! ⏳ Let me check the training status.

(Calling tool `check_gnn_training`)

Assistant

Training is in progress... 🔄 Let me check again in a moment.

(Calling tool `check_gnn_training`)

Assistant

Still training... ⏱️

(Calling tool `check_gnn_training`)

Assistant

Training in progress... 🔄

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Assistant

Excellent! 🎉 The GNN model training is complete with impressive results:

Metric	Value
Best Validation Accuracy	95.06%
Test Accuracy	94.61%

Now let me select a correctly predicted **active compound** from the test set.

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! Found a correctly predicted **active compound**:

- **SMILES**: `Clc1cccc(N2CCN(CCCCN3CCc4ncsc4CC3)CC2)c1Cl`
- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 99.99%

Now let me select a correctly predicted **inactive compound**.

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! Found a correctly predicted **inactive compound**:

- **SMILES**: `C#CCN(C)CC(=C)c1ccc(F)c(C)c1`
- **True Label**: 0 (Inactive)
- **Predicted Label**: 0 (Inactive)
- **Confidence**: 99.39%

Now let me explain both predictions using EdgeSHAPer. Starting with the **active compound**:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the **active compound** is complete:

- **Trustworthiness**: 79.49%
- **Fidelity**: 76.84%

Now let me explain the **inactive compound**:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the **inactive compound** is complete:

- **Trustworthiness**: 98.29%
- **Fidelity**: 96.79%

Now let me visualize both explanations:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the **inactive compound**:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me get the top bond environments for both compounds to provide detailed chemical interpretation:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Excellent! Now I have comprehensive EdgeSHAPer explanations for both compounds. Let me provide a detailed comparison and chemical interpretation:

📊 GNN Model Training Summary

Metric	Value
Architecture	GCN (Graph Convolutional Network)
Training Samples	6,010
Test Samples	835
Best Validation Accuracy	95.06%
Test Accuracy	94.61%
Training Epochs	100

🔬 Compound Comparison

Compound 1: Active (Predicted Class 1)

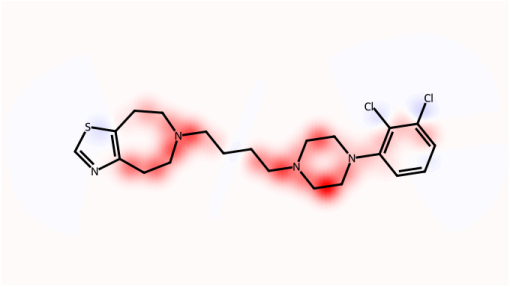
Property	Value
SMILES	`Clc1cccc(N2CCN(CCCCN3CCc4ncsc4CC3)CC2)c1Cl`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	99.99%
Trustworthiness	79.49%
Fidelity	76.84%
Number of Edges	62

Compound 2: Inactive (Predicted Class 0)

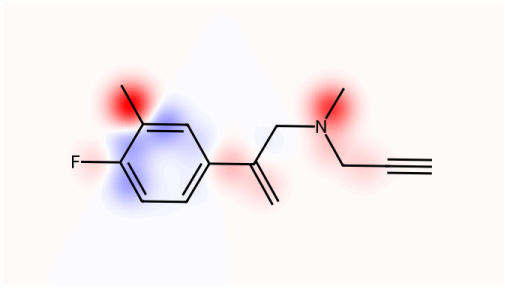
Property	Value
SMILES	`C#CCN(C)CC(=C)c1ccc(F)c(C)c1`
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	99.39%
Trustworthiness	98.29%
Fidelity	96.79%
Number of Edges	32

🖼️ EdgeSHAPer Visualizations

Active Compound - Edge Importance Heatmap



Inactive Compound - Edge Importance Heatmap



🔍 Key Differences & Chemical Interpretation

1. Molecular Complexity

Feature	Active Compound	Inactive Compound
Molecular Weight	Higher (bifunctional structure)	Lower (simpler structure)
Nitrogen Count	4 nitrogen atoms (bifunctional)	1 nitrogen atom (tertiary amine)
Aromatic Rings	2 (dichlorobenzene + benzothiazole)	1 (fluorotoluene)
Chlorine Atoms	2 chlorine atoms (Cl)	0 chlorine atoms
Sulfur Atoms	1 sulfur atom (thiazole ring)	0 sulfur atoms
Fluorine Atoms	0 fluorine atoms	1 fluorine atom (F)

2. Top Contributing Bonds (EdgeSHAPer Analysis)

Active Compound - Top 5 Positive (Activity-Promoting) Bonds:

Bond	Environment	Shapley Value	Interpretation
C-C (ring)	`NCCN` (piperazine linker)	+0.119	**Critical**: The piperazine nitrogen-carbon linkage strongly promotes activity
N-C (amine)	`CCN(C)C` (dialkylamine)	+0.085	**Strong**: Tertiary amine connectivity is crucial
C-N (amine)	`CCN(C)C` (dialkylamine)	+0.081	**Strong**: Nitrogen-centered bonding pattern
C-C (ring)	`NCCN` (piperazine linker)	+0.076	**Important**: Second piperazine ring bond
N-C (ring)	`CCN(C)C` (piperazine)	+0.076	**Important**: Piperazine nitrogen-carbon bond

Inactive Compound - Top 3 Bonds:

Bond	Environment	Shapley Value	Interpretation
C-C (aromatic)	`cc(c)C` (methyl-phenyl)	+0.145	**Promotes activity** (but compound is inactive overall)
N-C (amine)	`CN(C)C` (dialkylamine)	+0.108	**Promotes activity**
C-C (aromatic)	`ccc(c)C` (phenyl ring)	-0.061	**Depresses activity**

3. Critical Chemical Insights

What Makes the Active Compound Active:

1. **Bifunctional Structure**: The active compound has two distinct pharmacophores:
    - **Dichlorobenzene ring**: Provides hydrophobic/aromatic interactions

- **Benzothiazole ring**: Provides hydrogen bonding and π-π stacking capabilities
- **Piperazine linker**: Connects both rings, providing optimal spacing

2. **Multiple Nitrogen Centers**: 4 nitrogen atoms create multiple hydrogen bond acceptors/donors, enabling diverse target interactions
3. **Chlorine Substituents**: The two chlorine atoms on the benzene ring enhance lipophilicity and can participate in halogen bonding
4. **Extended Conjugation**: The two aromatic systems connected through flexible linkers allow the molecule to adopt multiple conformations for optimal binding

Why the Inactive Compound is Inactive:

1. **Simpler Structure**: Only one aromatic ring (fluorotoluene) limits interaction diversity
2. **Limited Nitrogen Content**: Only 1 nitrogen atom reduces hydrogen bonding potential
3. **Short Chain**: The alkyne (C#C) and short alkyl chains limit conformational flexibility
4. **Negative Aromatic Contributions**: The aromatic ring bonds have **negative** Shapley values (-0.061), meaning they actually **decrease** the activity prediction
5. **Insufficient Pharmacophores**: Lacks the key structural features (multiple aromatic rings, heteroatoms, optimal spacing) that characterize active compounds in this dataset

4. Summary of Key Differences

Aspect	Active	Inactive	Impact
Nitrogen atoms	4	1	Higher nitrogen count correlates with activity
Aromatic rings	2	1	Multiple rings enable better target binding
Halogens	Cl (2)	F (1)	Chlorine enhances lipophilicity
Sulfur	Yes (thiazole)	No	Sulfur heteroatom adds binding diversity
Linker length	Longer (piperazine)	Shorter (alkyne)	Optimal spacing is crucial
Bond contributions	Mostly positive	Mixed (negative aromatics)	Positive bonds drive activity

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📝 Conclusion

The GNN model successfully learned that **activity is driven by**:

1. **Multiple nitrogen-containing heterocycles** (especially piperazine and thiazole)
2. **Bifunctional architecture** with two aromatic systems
3. **Halogen substitution** (particularly chlorine)
4. **Extended conjugation** with flexible linkers

The **inactive compound**, despite having some activity-promoting features (like the tertiary amine), lacks the critical bifunctional structure and multiple heteroatoms needed for binding. Its aromatic bonds actually contribute negatively to the activity prediction, highlighting the importance of specific aromatic patterns rather than just any aromatic system.